

3.1. Phương pháp khử Gauss

0.1 Bai 3: Use Gaussian elimination with backward substitution and two-digit rounding arithmetic to solve the following linear systems. Do not reorder the equations. (The exact solution to each system is $x_1 = 1, x_2 = -1, x_3 = 3$.)

a.
$$\begin{cases} 4x_1 - x_2 + x_3 = 8 \\ 2x_1 + 5x_2 + 2x_3 = 3 \\ x_1 + 2x_2 + 4x_3 = 11 \end{cases}$$

b.
$$\begin{cases} 4x_1 + x_2 + 2x_3 = 9 \\ 2x_1 + 4x_2 - x_3 = -5 \\ x_1 + x_2 - 3x_3 = -9 \end{cases}$$

0.2 Bai 4: Use Gaussian elimination with backward substitution and two-digit rounding arithmetic to solve the following linear systems. Do not reorder the equations. (The exact solution to each system is $x_1 = -1, x_2 = 1, x_3 = 3$.)

a.
$$\begin{cases} -x_1 + 4x_2 + x_3 = 8 \\ \frac{5}{3}x_1 + \frac{2}{3}x_2 + \frac{2}{3}x_3 = 1 \\ 2x_1 + x_2 + 4x_3 = 11 \end{cases}$$

b.
$$\begin{cases} 4x_1 + 2x_2 - x_3 = -5 \\ \frac{1}{9}x_1 + \frac{1}{9}x_2 - \frac{1}{3}x_3 = -1 \\ x_1 + 4x_2 + 2x_3 = 9 \end{cases}$$

0.3 Bai 5: Use the Gaussian Elimination Algorithm to solve the following linear systems, if possible, and determine whether row interchanges are necessary:

a.
$$\begin{cases} x_1 - x_2 + 3x_3 = 2 \\ 3x_1 - 3x_2 + x_3 = -1 \\ x_1 + x_2 = 3 \end{cases} \quad \text{b.} \quad \begin{cases} 2x_1 - 1.5x_2 + 3x_3 = 1 \\ -x_1 + 2x_3 = 3 \\ 4x_1 - 4.5x_2 + 5x_3 = 1 \end{cases}$$

b.
$$\begin{cases} 2x_1 = 3 \\ x_1 + 1.5x_2 = 4.5 \\ -3x_2 + 0.5x_3 = -6.6 \\ 2x_1 - 2x_2 + x_3 + x_4 = 0.8 \end{cases}$$

c.
$$\begin{cases} x_1 + x_2 + x_4 = 2 \\ 2x_1 + x_2 - x_3 + x_4 = 1 \\ 4x_1 - x_2 - 2x_3 + 2x_4 = 0 \\ 3x_1 - x_2 - x_3 + 2x_4 = -3 \end{cases}$$

0.4 Bai 6: Use the Gaussian Elimination Algorithm to solve the following linear systems, if possible, and determine whether row interchanges are necessary:

$$\text{a. } \begin{cases} x_2 - 2x_3 = 4 \\ x_1 - 2x_2 + x_3 = 6 \\ x_1 - x_3 = 2 \end{cases} \quad \text{b. } \begin{cases} x_1 - \frac{1}{2}x_2 + x_3 = 4 \\ 2x_1 - x_2 - x_3 + x_4 = 5 \\ x_1 + x_2 + \frac{1}{3}x_3 = 2 \\ x_1 - \frac{1}{2}x_2 + x_3 + x_4 = 5 \end{cases}$$

$$\text{b. } \begin{cases} 2x_1 - x_2 + x_3 - x_4 = 6 \\ x_2 - x_3 + x_4 = 5 \\ x_4 = 5 \\ x_3 - x_4 = 3 \end{cases} \quad \text{d. } \begin{cases} x_1 + x_2 + x_4 = 2 \\ 2x_1 + x_2 - x_3 + x_4 = 1 \\ -x_1 + 2x_2 + 3x_3 - x_4 = 4 \\ 3x_1 - x_2 - x_3 + 2x_4 = -3 \end{cases}$$